

AKSHATHA DUNGI

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OBJECTIVE

Aspiring Software Engineer, published IEEE research on neural network calibration, seeking full-time backend or full-stack engineering with AI-integration roles where I can own and ship end to end.

EDUCATION

Bachelor of Technology in Computer Science and Systems Engineering

Andhra University College of Engineering, Visakhapatnam

2022 – 2026

CGPA: 9.07/10

- **Achievement:** Academic Excellence Award – Highest CGPA in CSE Department (2026)
- **Coursework:** Data Structures, Algorithms, Databases, Computer Networks, Operating Systems, Object-Oriented Programming, Machine Learning
- **Publication:** "*AI Confidence Calibration Failure Detection*" – IEEE 2026, co-author research on confidence calibration across 5 neural network architectures

SKILLS

Languages	C++, Python, JavaScript, TypeScript, SQL, Java
Frontend	React, HTML5, CSS3, Tailwind CSS, Responsive Design
Backend	FastAPI, Node.js, Express.js, RESTful APIs, JWT Authentication, SSE (Server-Sent Events)
AI/ML	TensorFlow, PyTorch, Scikit-learn, LLM Integration (Gemini API, OpenAI API), RAG, Vector Search (pgvector, FAISS)
Databases	PostgreSQL, MySQL, MongoDB, SQLite
Cloud/DevOps	Docker, Git, GitHub Actions, CI/CD, Vercel, Railway

EXPERIENCE

Summer Research Intern, ISI Kolkata.

Jun 2025 – Aug 2025

Engineered an Artificial Neural Network (ANN) achieving 97% multi-class classification accuracy, improving predictive performance by 15% over existing baseline models. Implemented model evaluation metrics and visualization pipelines for stakeholder reporting, using Python, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, and Matplotlib.

PROJECTS

Remi – AI Chatbot Widget

([GitHub](#))

- Built and deployed a full-stack, embeddable AI chatbot widget using React, TypeScript, and FastAPI, featuring JWT authentication, SSE streaming, and RAG-backed document Q&A over PDF, DOCX, and XLSX uploads
- Designed a retrieval pipeline using Gemini embeddings with pgvector cosine-similarity search in PostgreSQL, including OCR support for scanned documents via PyMuPDF
- Implemented production security hardening (rate limiting, prompt-injection sanitization, audit logging) and maintained 191 passing tests with CI/CD via GitHub Actions
- Deployed live on Vercel (frontend) and Railway (backend) with PostgreSQL persistence

AI Confidence Calibration Failure Detection

(Published – IEEE 2026)

- First architecture-comparative study on confidence calibration failure across 5 neural networks (MLP, Deep MLP, Residual MLP, RNN, LSTM) on the FinSen financial dataset (160K records)
- Achieved 0.0032 ECE (MLP) and 0.983 accuracy across 240 experiments; discovered self-recovering calibration behavior in Residual MLP, a previously unreported finding

- Established Gradient-ECE correlation ($r=0.568$) as a novel early-warning signal for calibration failure
- Tech stack: PyTorch, NumPy, Pandas, Matplotlib, Scikit-learn, Jupyter, Docker

LEADERSHIP

- **Research Team Lead, Andhra University** – Led a 5-person cross-functional research team on a 6-month AI calibration study, coordinating model implementation, data pipeline development, and results publication across the group.
- **Intern, a-hub Incubation Center** – Co-organized a campus internship drive from scratch, sourcing and onboarding 10+ corporate partners and directly facilitating 30+ student placements across Business, Engineering, and IT tracks.